

Lesson 1: Solid, Liquid and Gas

Module 1: Properties of Matter

Learning Objectives:

- Define matter as anything that has mass and occupies space.
- Describe the shape and volume of solids, liquids, and gases.
- Relate the arrangement and movement of particles to each state of matter.
- Classify familiar objects as solid, liquid, or gas.

Particle Arrangement in the Three Common States of Matter

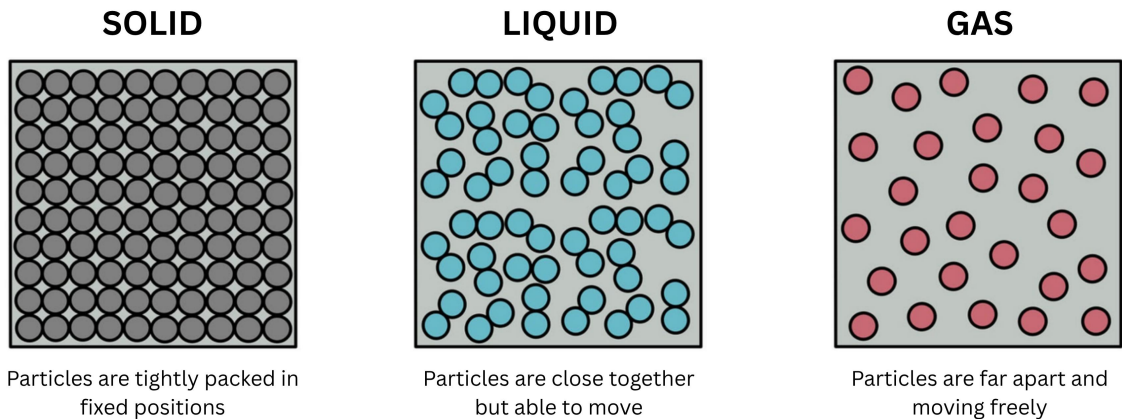


Figure 1. Particle arrangement in solids, liquids, and gases.

Matter

Matter is anything that has mass and occupies space. A book, water, air, a spoon, and a balloon all contain matter. Matter can be described using observable properties such as shape, volume, texture, color, and how it behaves inside a container.

1. Solid — A solid has a definite shape and a definite volume. Its particles are packed closely together and mainly vibrate in fixed positions. Examples include a stone, pencil, wooden block, and ice cube.

2. Liquid— A liquid has a definite volume but no definite shape. It flows and takes the shape of its container. Its particles are close together but can slide past one another. Examples include water, milk, and cooking oil.

3. Gas— A gas has no definite shape and no definite volume. It spreads out and fills the available space in its container. Its particles are far apart and move freely. Air and water vapor are examples of gases.

Comparing the Three States

- **Shape:** Solids keep their shape; liquids and gases take the shape of their container.
- **Volume:** Solids and liquids have definite volume; gases expand to fill available space.
- **Particle arrangement:** Solid particles are tightly packed, liquid particles are close but mobile, and gas particles are widely spaced.
- **Flow:** Liquids and gases can flow, while solids normally do not.

Everyday Examples

An ice cube is a solid, melted ice is liquid water, and water vapor is a gas. These examples show that the same substance can exist in different states. Heating and cooling can cause changes of state, but understanding each state should come first.